

TrES-1b

R-1664

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_190821 · created 2026-08-02T09:55:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458716.6808 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.013
Transit depth (Rp/Rs)^2	2.61 +/- 0.42 [%]
Orbital Inclination (inc)	86.94 +/- 1.02
Airmass coefficient 1 (a1)	1.149 +/- 0.012
Airmass coefficient 2 (a2)	-0.13 +/- 0.074
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	None
Optimal Aperture	4.52
Optimal Annulus	12.81
Transit Duration (day)	0.0786 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.013 vs 0.1358 (deviation 1.64σ)
Duration fit vs published	0.0786 d vs 0.1042 d (deviation 3.27σ)
Mid-time O–C	-5.6 min
Depth SNR	10.5
Residual scatter	1.02 %

Light curve

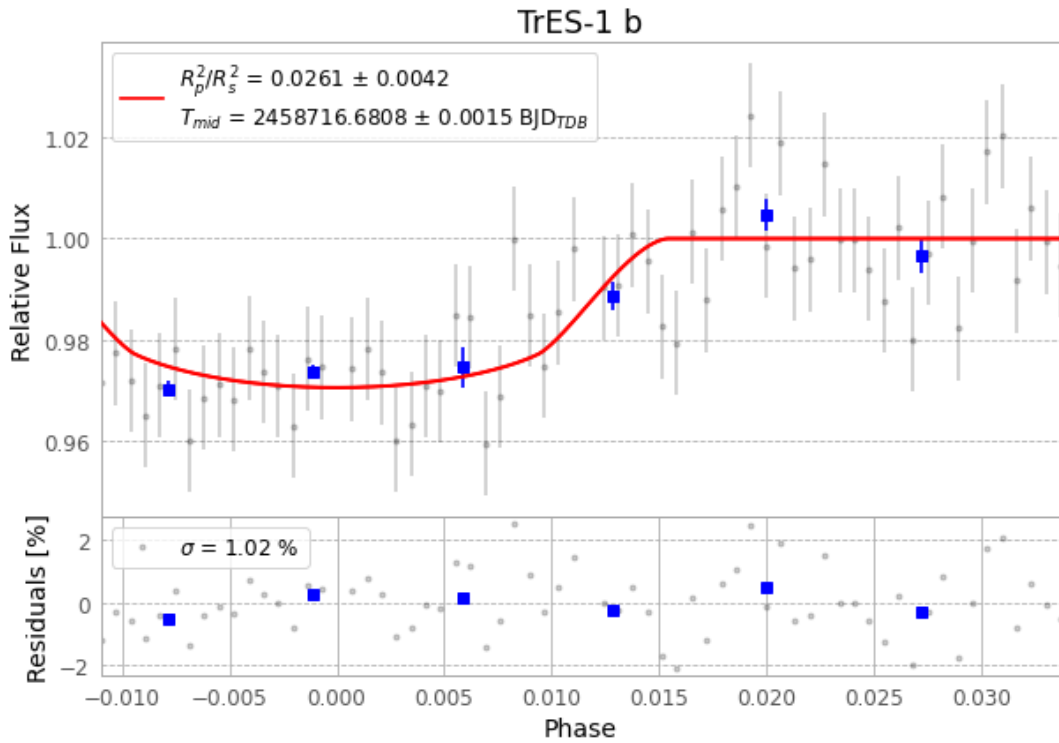


Figure 1 — FinalLightCurve_TrES-1 b_2019-08-20.png (SHA-256 44b20ce749a79168...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 174], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c060ed90ff7ee9f8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-1190821032712.FITS → TRES-1190821064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190821.log (3a6cf79ca3fc...), FinalLightCurve_TrES-1 b_2019-08-20.png (44b20ce749a7...), FinalLightCurve_TrES-1 b_2019-08-20.csv (474ae9c96606...)

Compute resources

Wall time 150.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 09:55Z.